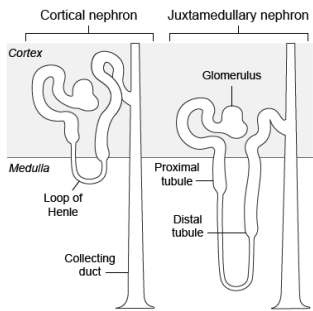


Camels inhabit water-deficient environments and exhibit an evolutionary adaptation that allows them to absorb more water than their ancestors that dwelled in water-rich environments. The diagram below shows the structure of two nephron types present in the kidneys of camels and other mammals.

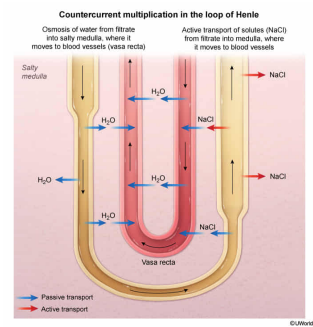


How would the number of cortical nephrons most likely differ from the number of juxtamedullary nephrons in camels and their ancestors? Camels would be expected to have:

- ☐ A. a lower number of juxtamedullary nephrons and a higher number of cortical nephrons compared to their ancestors. [23%]
- ☒ B. a higher number of juxtamedullary nephrons and a lower number of cortical nephrons compared to their ancestors. [70%]
- ☐ C. a similar number of juxtamedullary nephrons and a lower number of cortical nephrons compared to their ancestors. [2%]
- ☐ D. a lower number of juxtamedullary nephrons and a similar number of cortical nephrons compared to their ancestors. [2%]

Correct
71% answered correctly

Explanation:



Mammalian kidneys function to excrete waste and maintain appropriate solute and water concentrations throughout the body. Nephrons are the functional units of the kidney that filter blood and selectively secrete or reabsorb the contents of the resulting filtrate.

The kidneys are anatomically divided into an outer cortex and an inner "salty" medulla. In the cortex, glomerular capillaries filter blood into the Bowman's capsule of the nephron. This filtrate then moves through the proximal tubule and flows into the loop of Henle, a structure, divided into two limbs, that reabsorbs water and salt (NaCl). Countercurrent multiplication forms a concentration gradient within the loop of Henle that maximizes water reabsorption:

1. The descending limb, extending from the cortex into the salty medulla, is highly permeable to water but impermeable to NaCl. Consequently, the filtrate becomes more concentrated as water is passively reabsorbed via osmosis into the medulla, where it is taken up by blood vessels.
2. The ascending limb, traveling from the loop's lowest point in the medulla back toward the cortex, is impermeable to water but permeable to NaCl. NaCl is first passively reabsorbed into the medulla as the filtrate travels up the ascending limb. Yet as the limb nears the cortex, NaCl is actively transported out of the filtrate and into the medulla, maintaining the medulla's high salt concentration while facilitating continued water reabsorption in the descending limb.

Camels inhabit water-deficient environments and can reabsorb more water than their ancestors that inhabited water-rich environments. As the environment changed from water-abundant to water-deficient, these mammals likely physiologically evolved to maximize water absorption when they encountered a rare water source.

In the question diagram, juxtamedullary nephrons have long loops of Henle extending deep into the salty medulla to maximize water reabsorption; in contrast, cortical nephrons have shorter loops of Henle extending shorter distances into the medulla, making them less efficient at water reabsorption. Therefore, compared to their ancestors, camels have likely evolved a higher number of juxtamedullary nephrons and a lower number of cortical nephrons (Choices A, C, and D).

Educational objective:

In the loop of Henle, the active transport of NaCl from the ascending limb maintains the high salt concentration in the kidney's medulla, facilitating the reabsorption of water from the descending limb.

Time spent:0 sec
Subject:
Biology
QID:402349
System:
Digestion and Excretion